

Homework 2

Textbook reading

Please read Section 2.1.1 of the textbook, stopping when you get to Section 2.2.

Homework questions

For each of the following questions, explain how to count the described things. Most of the questions come from Durrett (our textbook).

- Give your method for counting, following the model of my example solutions from Homework 1.
- If any of the numbers you use in your counting match Durrett's special notations $P_{n,k}$ or $C_{n,k}$, give those notations, as in my earlier example solutions.
- I did the counting for you in some of the exercises. Your main job throughout is to explain the reasoning behind the counting.

1. How many five-letter "words" can you make if no letter is used twice and each word must contain at least one vowel (A, E, I, O, or U)? (5,451,720)

Here is a technique you can use for counting:

- Count the number of "words" without repeated letters but with no requirement about vowels.
- Count the number of "words" without repeated letters that have no vowels.
- The number you want ("at least aone vowel") is the difference between these two.

When you write your solution, explain the steps of this method. We will use this method often in situations involving "at least."

Number of words without repeated letters (no requirement about vowels):

These words are permutations of five letters taken from a set of 26 letters. There are $P_{26,5}$ of them.

Number of words without repeated letters but with no vowels:

These words are permutations of five letters taken from a set of 21 letters (26 letters minus the five vowels). There are $P_{21,5}$ of them.

Number of words with at least one vowel:

These are the words from the first group that are not in the second group. There are $P_{26,5} - P_{21,5}$ of them. That works out to be 5,451,720, as I promised.

2. How many ways can the letters in the following words be arranged? Please explain using the new method from August 21st (and Section 2.1.1).

money, banana, statistics

money: since the five letters are distinct, a rearrangement of this word is a permutation of the five letters. There are $P_{5,5}$ or $5!$ such permutations.

banana: we think of six spots where we place the six letters (one b, three a's, and two n's). We pick one spot for a b, a group of three spots for a's, and a group of two spots for n's. There are $6!/(1!3!2!) = 60$ ways to do that.

statistics: we think of ten spots where we place the ten letters (three s's, three t's, one a, two i's, and one c). We pick a group of three spots for s's, a group of three spots for t's, a spot for the a, a group of two spots for i's, and a spot for the c. There are $10!/(3!3!1!2!1!) = 50,400$ ways to do that.

3. A club with 50 members will form two committees, one with 8 members and the other with 7. How many ways can this be done if the committees must be disjoint (having no members in common)?

The value is 14,484,088,315,897,200. This number is so large that a regular calculator may not be able to show the exact value, though it may give you an approximate value in scientific notation. Explain how to get this value in two ways. First, pick the two committees in stages (the eight-member committee and then the seven-member), counting using the multiplication rule. Then explain how to count using the new method from August 21st (and Section 2.1.1).

First method:

We first choose a group of eight members to form the first committee. There are $C_{50,8}$ ways to do that. No matter which eight members we pick for the first committee, there are 42 left to pick from for the second committee. We pick seven of these to form the second committee. There are $C_{42,7}$ ways to do that. By the multiplication rule, there are $C_{50,8} C_{42,7}$ total ways to choose the two committees.

Second method:

We divide the 50 members into three groups, a group of eight for the first committee, a group of seven for the second committee, and a group of 35 who will not be on either committee. The multinomial counting method from Durrett tells us that there are $50!/(8! 7! 35!)$ ways to do that.

4. The same club will form two committees (eight and seven members again), but now the committees can overlap. How many ways can this be done? The value is 53,625,801,828,060,000. You may not be able to see that exact value on a calculator. Explain how to count it.

We use the multiplication rule. First we pick eight members for the first committee. There are $C_{50,8}$ ways to do that. Then we pick seven members (out of all 50) for the second committee. There are $C_{50,7}$ ways to do that. By the multiplication rule, there are $C_{50,8} C_{50,7}$ total possibilities.

5. How many ways can 5 history books, 3 math books, and 4 novels be arranged on a shelf if the books of each type must be together? (103,680)

We use the multiplication rule. First we decide how to put the categories (history, math, novels) in order on the shelf. There are $P_{3,3}$ or $3!$ ways to order the categories. We then put the history books in order. There are $5!$ ways to do that. Next we put the math books in order. There are $3!$ ways to do that. And finally we

put the novels in order. There are $4!$ ways to do that. By the multiplication rule, there are $3!5!4!3!$ total ways to arrange the books, which works out to be 103,680, as promised.

6. Look at the class worksheet from August 21st to review what a poker hand is and when a poker hand is a straight. How many straights are there where the lowest card is the ace of hearts?

We use the multiplication rule. We can make a straight starting with the ace of hearts in order. The second card has value 2 and can be from any suit. There are four possibilities. The third card has value 3 and can be from any suit (four possibilities). The fourth card has value 4 and can be from any suit (four possibilities). The fifth card has value 5 and can be from any suit (four possibilities). By the multiplication rule, there are $(4)(4)(4)(4) = 256$ possible straights starting with the ace of hearts.